

5(a)

glycine in monomer column only ;

5

monomer	polymer	monosaccharide	polysaccharide
thymine (nucleotide) } α -glucose } ; β -glucose }	cellulose messenger RNA } glycogen } ; α -globin } 1 glycine	β -glucose } α -glucose } ; 1 glycine	cellulose } glycogen } ; 1 glycine

1 incorrect spellings

if α -globin appears once and is in monomer column instead of polymer column, and the monomer and polymer columns are otherwise correct, allow one mark (ecf)

5(b)

any three from:

3

allow points from an annotated diagram

1 phosphate head = hydrophilic / polar ;

2 fatty acid / hydrocarbon, tail / chains = hydrophobic / non-polar ;

if mps 1 and 2 not gained, allow one mark for, hydrophilic / polar, heads and, hydrophobic / non-polar, tails

3 (so) heads face, watery environment / tissue fluid / cytoplasm / cytosol / aqueous environment ;

4 (fatty acid) tails, form hydrophobic core / form area away from water / face each other / AW ;

5 ref. tails and hydrophobic interactions ;